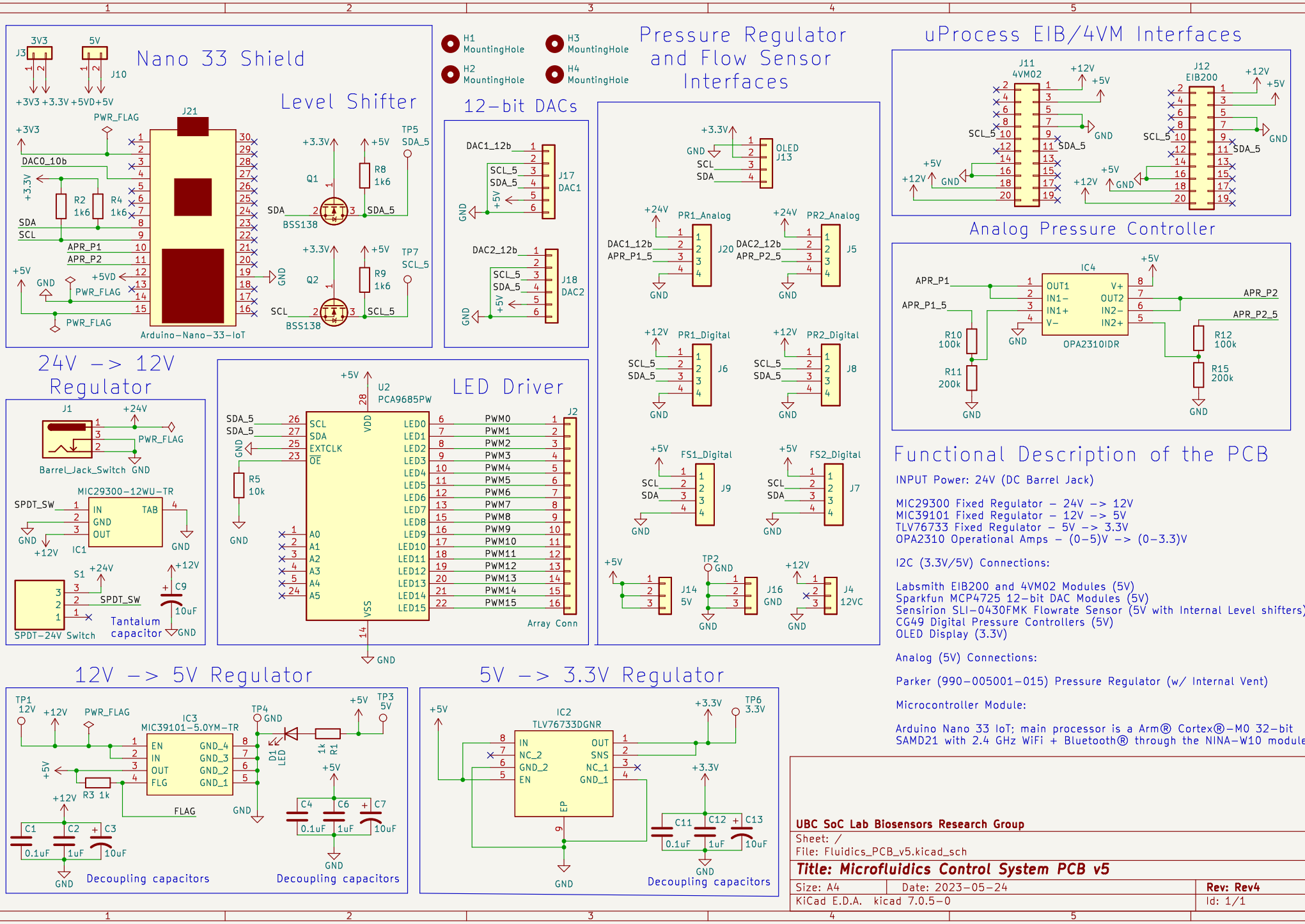


Functional Description of the PCB

INPUT Power: 24V (DC Barrel Jack)

MIC29300 Fixed Regulator – 24V -> 12V
MIC39101 Fixed Regulator – 12V -> 5V
TLV76733 Fixed Regulator – 5V -> 3.3V
OPA2310 Operational Amps – (0-5)V -> (0-3.3)V

I2C (3.3V/5V) Connections:

Labsmith EIB200 and 4VM02 Modules (5V)
Sparkfun MCP4725 12-bit DAC Modules (5V)
Sensirion SLI-0430FMK Flowrate Sensor (5V with Internal Level shifters)
CG49 Digital Pressure Controllers (5V)
OLED Display (3.3V)

Analog (5V) Connections:

Parker (990-005001-015) Pressure Regulator (w/ Internal Vent)

Microcontroller Module:

Arduino Nano 33 IoT; main processor is a Arm® Cortex®-M0 32-bit SAMD21 with 2.4 GHz WiFi + Bluetooth® through the NINA-W10 module

UBC SoC Lab Biosensors Research Group

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Title: Microfluidics Control System PCB v5

Size: A4	Date: 2023-05-24	**Rev: Rev4**
KiCad E.D.A. kicad 7.0.5-0		Id: 1/1